Intervals of Convergence with Endpoint Checks

Answer Key

AP Calculus BC

Quick Answers

1. $\frac{1}{2}$
2. 2, 6
3. $-\log(2)$
4. ∞
5. $\frac{1}{5}$
6. $\frac{\pi^2}{6}$, $-\frac{\pi^2}{12}$
7. 0, e
8. $\frac{1}{3}$

Curriculum

Level: AP Calculus BC
LIM-7, LIM-8 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 12 tagged checks.

1. Sensor model $C(x) = \sum_{n\geq 1}(x-4)^n/(n\cdot 2^n)$: ratio test and radius.

With $a_n = (x-4)^n/(n\cdot 2^n)$,

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(x-4)^{n+1}}{(n+1)\,2^{n+1}}\cdot\frac{n\,2^n}{(x-4)^n}\right| = |x-4|\cdot\frac{n}{2(n+1)}.$$

The x-free factor is what the ratio test needs in the limit. Dividing numerator and denominator by n gives $\dfrac{1}{2(1+1/n)}$, and $1/n \to 0$.

The limit is $\boxed{L = \frac{1}{2}}$

The ratio test demands $|x-4|\cdot L < 1$, i.e. $|x-4| < 1/L$, so the radius is $R = 1/L$ bars. $\boxed{R = 2 \text{ bars}}$

2. Endpoints from the center and radius.

The series is centered at $c = 4$ bars with $R = 2$ bars, and the ratio test guarantees convergence exactly on $|x-4| < 2$. The boundary of that set is where $|x-4| = 2$, which is the two points $c-R$ and $c+R$.

Left endpoint: $c - R = 4 - 2$. $\boxed{x = 2 \text{ bars}}$

Right endpoint: $c + R = 4 + 2$. $\boxed{x = 6 \text{ bars}}$

Note: the ratio test says nothing at either point — there $|x-4|\cdot L = 1$, and the test is inconclusive. That is why problems 3 and 4 exist.

3. Left endpoint $x = 2$ bars.

Substitute $x = 2$, so $x - 4 = -2$:

$$\sum_{n=1}^{\infty} \frac{(-2)^n}{n \cdot 2^n} = \sum_{n=1}^{\infty} \frac{(-1)^n 2^n}{n \cdot 2^n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}.$$

This is the alternating harmonic series. **Test:** the alternating series test — the terms $1/n$ are positive, decreasing, and tend to 0 — so it converges. (It converges conditionally: $\sum 1/n$ itself diverges.)

Its exact sum is the value of the Maclaurin series $\ln(1+u) = \sum_{n\geq 1}(-1)^{n-1}u^n/n$ at $u = 1$, with the sign flipped because our numerator is $(-1)^n$, not $(-1)^{n-1}$.

$$\boxed{-\ln 2}$$

So $x = 2$ bars *belongs* to the interval, and it earns a square bracket.

4. Right endpoint $x = 6$ bars, and the full interval.

Substitute $x = 6$, so $x - 4 = 2$:

$$\sum_{n=1}^{\infty} \frac{2^n}{n \cdot 2^n} = \sum_{n=1}^{\infty} \frac{1}{n},$$

the harmonic series. **Test:** the p-series test with $p = 1$; a p-series diverges for $p \leq 1$, so this endpoint diverges and is *excluded*. (The nth-term test is useless here: $1/n \to 0$, yet the series still diverges.)

Combining problems 1–4: the ratio test gives convergence on $2 < x < 6$, the left endpoint converges and the right endpoint diverges.

$$\boxed{2 \leq x < 6}$$

The model is usable for pressures from 2 bars up to but not including 6 bars.

5. Gain model $G(x) = \sum_{n\geq 1}(x+3)^n/(n^2 \cdot 5^n)$: ratio test and radius.

With $a_n = (x+3)^n/(n^2 \cdot 5^n)$,

$$\left|\frac{a_{n+1}}{a_n}\right| = |x+3| \cdot \frac{n^2}{5\,(n+1)^2}.$$

Divide numerator and denominator by n^2: the factor becomes $\dfrac{1}{5(1+1/n)^2} \to \dfrac{1}{5}$.

The limit is $\boxed{L = \frac{1}{5}}$

Then $|x+3| < 1/L$, so the radius is $R = 1/L$ volts.

$$\boxed{R = 5 \text{ volts}}$$

6. Both endpoints of the gain model, and the full interval.

The center is $c = -3$ volts and $R = 5$ volts, so the endpoints are $-3+5 = 2$ and $-3-5 = -8$ volts.

At $x = 2$ volts $(x + 3 = 5)$:

$$\sum_{n=1}^{\infty} \frac{5^n}{n^2 \cdot 5^n} = \sum_{n=1}^{\infty} \frac{1}{n^2},$$

a p-series with $p = 2 > 1$, so it converges. Its exact sum is the Basel value.

$$\boxed{\frac{\pi^2}{6}}$$

At $x = -8$ volts $(x + 3 = -5)$:

$$\sum_{n=1}^{\infty} \frac{(-5)^n}{n^2 \cdot 5^n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2},$$

which converges absolutely by comparison with $\sum 1/n^2$ (the alternating series test also settles it). Its exact sum is the negative of the alternating Basel value, $-\eta(2) = -\frac{1}{2} \cdot \frac{\pi^2}{6}$.

$$\boxed{-\frac{\pi^2}{12}}$$

Both endpoints converge, so both earn square brackets and the interval is $-8 \le x \le 2$ volts.

7. Reaction-rate model $R(x) = \sum_{n\ge0}(x-2)^n/n!$.

(a) With $a_n = (x-2)^n/n!$,

$$\left|\frac{a_{n+1}}{a_n}\right| = |x-2| \cdot \frac{n!}{(n+1)!} = |x-2| \cdot \frac{1}{n+1},$$

and $1/(n+1) \to 0$ as $n \to \infty$.
The limit is $\boxed{L = 0}$

(b) Because $L = 0$, the product $|x-2| \cdot L = 0 < 1$ for *every* real x, so the radius is $R = \infty$ and the interval of convergence is all real numbers, written $(-\infty, \infty)$. There are no endpoints to test: an infinite radius means the boundary set $|x-2| = R$ is empty, so the inconclusive case of the ratio test never arises.

(c) Setting $x = 3$ gives $x - 2 = 1$, so $R(3) = \sum_{n\ge0} 1/n!$, which is the Maclaurin series for e^u at $u = 1$.

$$\boxed{R(3) = e}$$

8. Energy model $E(x) = \sum_{n\geq 1} n\,(x-1)^n/3^n$.

(a) With $a_n = n\,(x-1)^n/3^n$,

$$\left|\frac{a_{n+1}}{a_n}\right| = |x-1| \cdot \frac{n+1}{3n},$$

and dividing through by n gives $\dfrac{1+1/n}{3} \to \dfrac{1}{3}$.

The limit is $\boxed{L = \frac{1}{3}}$

So $|x-1| < 3$, the radius is $R = 3$, and the endpoints are $1-3 = -2$ and $1+3 = 4$.

(b) *(Open reasoning — flagged for manual review.)* **Model answer.** At $x = 4$ the terms become $n \cdot 3^n/3^n = n$, so the series is $\sum n$, whose terms grow without bound; since the terms do not approach 0, the nth-term test for divergence applies and the series diverges. At $x = -2$ the terms become $n(-3)^n/3^n = (-1)^n n$, whose absolute value n again does not approach 0; the nth-term test applies here too and the series diverges. The alternating series test does *not* rescue this endpoint, because that test requires terms decreasing to 0 and these increase without bound.

(c) Both endpoints diverge, so both are excluded and both brackets are round.

$\boxed{-2 < x < 4}$

Grading note: full credit requires (i) the correct simplification at each endpoint, (ii) the nth-term test named at both — not the alternating series test at $x = -2$ — and (iii) round brackets justified by the divergence, not asserted. A student who writes $[-2, 4)$ has tested only one endpoint.